

Chapter 3: Phase-Shift Control of the Dual Active Bridge (DAB)

Steady-State Operation of the Dual Active Bridge

*Modeling and Optimization of Bidirectional
Dual Active Bridge DC-DC Converter Topologies*

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The Lossless DAB Model

The DAB converter consists of two voltage-source full bridges interconnected through a high-frequency transformer and a series inductor L .

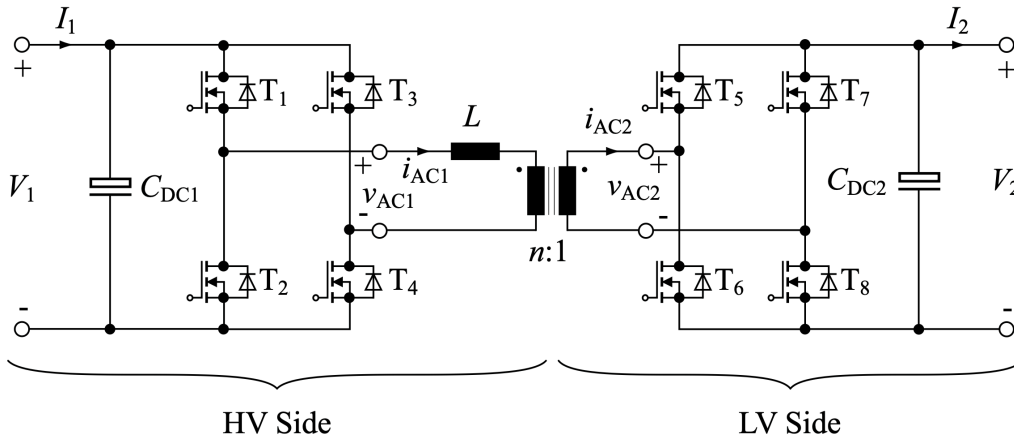


Figure 1: The DAB topology

Reducing the full topology to an analytically tractable equivalent circuit takes two steps.

First, each full bridge becomes an equivalent voltage source: seen from outside, a full bridge behaves as a square-wave voltage source. Second, the transformer is referred to the HV side. What remains is an extremely compact **single-loop model**.

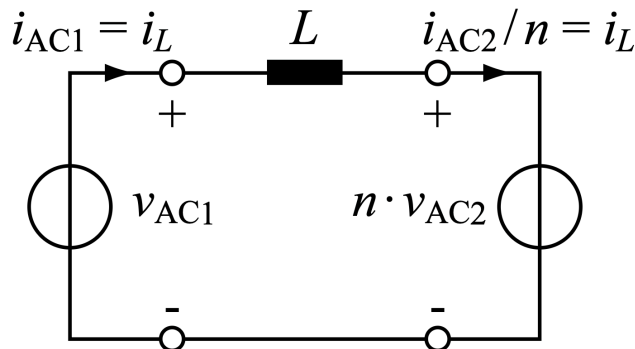


Figure 2: The lossless DAB equivalent model

$$v_R(t) = v_{AC1}(t) - n v_{AC2}(t) \quad (1)$$

Here v_R is the “net voltage” that actually drives the change of inductor current. The model rests on four idealizing assumptions: all losses are neglected; transformer magnetizing inductance and parasitic capacitances are neglected; all LV-side quantities are referred to the HV side; and V_1, V_2 are treated as constants.

The Four Switching States of the HV Full Bridge

The HV full bridge outputs **three voltage levels** ($+V_1, 0, -V_1$) through four switch combinations:

State	Conducting switches	v_{AC1}	Nature
I	T_1, T_4	$+V_1$	Forward drive
II	T_1, T_3	0	Freewheeling
III	T_2, T_4	0	Freewheeling
IV	T_2, T_3	$-V_1$	Reverse drive

Table 1: All switching states of the HV full bridge. The LV full bridge is fully symmetric (T_5 – T_8 , producing $\pm V_2$ or 0).

The upper limit of D is $1/2$. A DAB produces a bipolar square wave: positive for one half-period, negative for the other. A positive pulse can at most fill one half-period, hence $D_{\max} = 1/2$. If $D > 1/2$, the positive pulse intrudes into the negative half-period $\rightarrow$ average value of v_{AC} is no longer zero $\rightarrow$ DC bias appears in the transformer $\rightarrow$ core saturates $\rightarrow$ converter blows up. This differs from the usual buck-converter convention $D \in (0, 1)$.

Single Phase Shift Modulation (SPS)

SPS is the most classical and simplest modulation scheme for a DAB (De Doncker, 1991). Its core constraint is $D_1 = D_2 = 1/2$, leaving only φ adjustable. Consequently $v_{AC1}(t) \in \{+V_1, -V_1\}$ and $v_{AC2}(t) \in \{+V_2, -V_2\}$ — **voltage never rests at 0, and no freewheeling state exists**. Both full bridges toggle only between states I and IV; states II and III never occur.

Half-Period Antisymmetry

In steady state, the square waves and inductor current satisfy:

$$v_{AC1}(t + T_S/2) = -v_{AC1}(t), \quad v_{AC2}(t + T_S/2) = -v_{AC2}(t), \quad i_L(t + T_S/2) = -i_L(t) \quad (2)$$

Only the first half-period $[0, T_S/2]$ therefore needs analysis; the second half-period is its sign-inverted copy. In particular, the steady-state boundary condition is $i_L(T_S/2) = -i_L(0) \equiv -i_{L,0}$.

Switching-State Table over a Period

Assume $\varphi > 0$ (HV leads LV, power flows HV $\rightarrow$ LV); the phase shift corresponds to time $T_\varphi = \varphi/(2\pi f_S)$.

Over one period v_R is strictly antisymmetric: $\{V_1 + nV_2, V_1 - nV_2, -(V_1 + nV_2), -(V_1 - nV_2)\}$. Positive and negative values pair up, guaranteeing $\int_0^{T_S} v_R dt = 0$, i.e. **no DC bias in the transformer**.

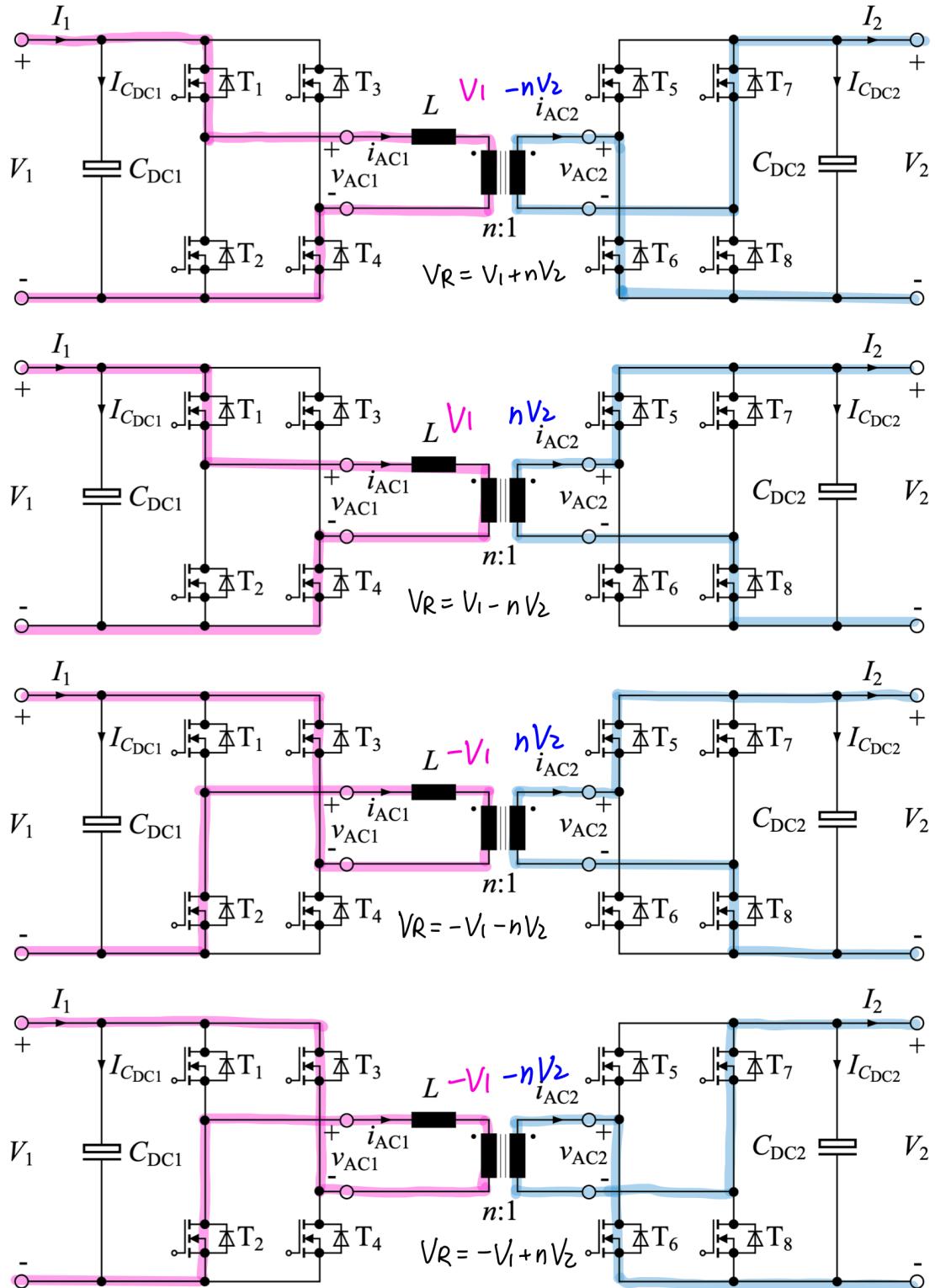


Figure 3: Energy flow over one period under SPS

Int.	Time	HV ON	LV ON	v_{AC1}	nv_{AC2}	v_R	i_L trend
I	$0 \rightarrow T_\varphi$	T_1, T_4	T_6, T_7	$+V_1$	$-nV_2$	$+(V_1+nV_2)$ large pos.	steep rise
II	$T_\varphi \rightarrow T_S/2$	T_1, T_4	T_5, T_8	$+V_1$	$+nV_2$	$+(V_1-nV_2)$ small	slow change
III	$T_S/2 \rightarrow T_S/2+T_\varphi$	T_2, T_3	T_5, T_8	$-V_1$	$+nV_2$	$-(V_1+nV_2)$ large neg.	steep fall
IV	$T_S/2+T_\varphi \rightarrow T_S$	T_2, T_3	T_6, T_7	$-V_1$	$-nV_2$	$-(V_1-nV_2)$ small	slow change

Table 2: Complete switching states over one period under SPS modulation

Derivation of the SPS Power

Step 1: Simplifying the power integral (3.10). By half-period antisymmetry, $p_1(t)$ is identical over both half-periods (the two minus signs cancel: $(-v_{AC1})(-i_L) = v_{AC1}i_L$), hence:

$$P_1 = \frac{2}{T_S} \int_0^{T_S/2} v_{AC1}(t) i_L(t) dt \quad (3)$$

Over the first half-period $v_{AC1}(t) = +V_1$ is constant and pulls out of the integral:

$$P_1 = \frac{2V_1}{T_S} \int_0^{T_S/2} i_L(t) dt \quad (4)$$

The problem reduces to **computing the area under i_L over $[0, T_S/2]$** .

Step 2: Piecewise expression for inductor current. In interval I, $v_R = V_1 + nV_2$ (constant), and integrating gives:

$$i_L(t) = i_{L,0} + \frac{V_1 + nV_2}{L} t \quad \forall 0 < t < T_\varphi \quad (5)$$

Write $i_\varphi \equiv i_L(T_\varphi) = i_{L,0} + (V_1 + nV_2)T_\varphi/L$. In interval II, $v_R = V_1 - nV_2$ (constant):

$$i_L(t) = i_\varphi + \frac{V_1 - nV_2}{L} (t - T_\varphi) \quad \forall T_\varphi < t < T_S/2 \quad (6)$$

Step 3: Finding $i_{L,0}$ from half-period symmetry. Substituting $t = T_S/2$ into (3.11b) gives $i_L(T_S/2)$; imposing the steady-state condition $i_L(T_S/2) = -i_{L,0}$ then yields the initial current.

Expanding $i_L(T_S/2)$:

$$i_L(T_S/2) = i_{L,0} + \frac{V_1 + nV_2}{L} T_\varphi + \frac{V_1 - nV_2}{L} \left(\frac{T_S}{2} - T_\varphi \right) \quad (7)$$

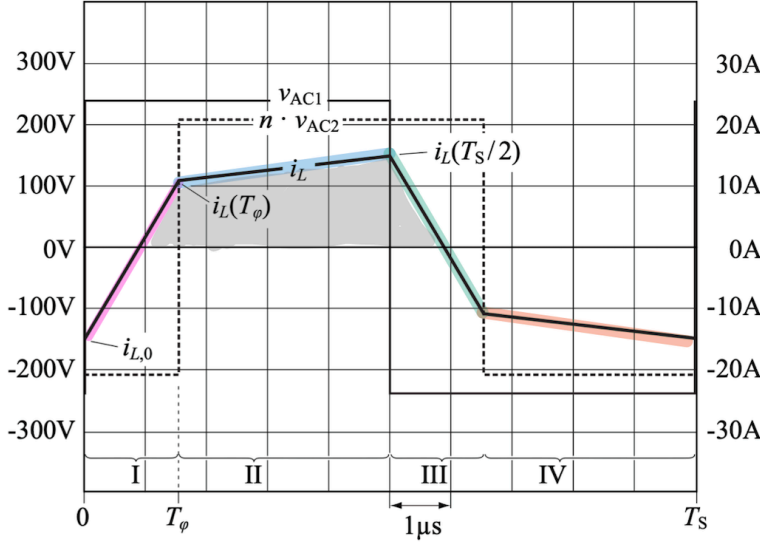
Setting $i_L(T_S/2) = -i_{L,0}$, i.e. $-2i_{L,0} = [\dots]$, gives:

$$i_{L,0} = \frac{\pi(nV_2 - V_1) - 2\varphi nV_2}{4\pi f_S L} \quad (8)$$

(where $T_S = 1/f_S$ and $T_\varphi = \varphi/(2\pi f_S)$ have been substituted).

Step 4: Computing $\int_0^{T_S/2} i_L dt$ by the trapezoid-area method. Since i_L consists of two straight-line segments, the integral is just the sum of two trapezoid areas. The three endpoint values are $i_L(0) = i_{L,0}$, $i_L(T_\varphi) = i_\varphi$, and $i_L(T_S/2) = -i_{L,0}$ (using the symmetry condition).

Trapezoid 1 (interval I): $S_1 = \frac{(i_{L,0} + i_\varphi) T_\varphi}{2}$



$$\text{slope 1: } V_1 + nV_2 / t \cdot L$$

$$\text{slope 2: } V_1 - nV_2 / t \cdot L$$

$$\text{slope 3: } -V_1 - nV_2 / t \cdot L$$

$$\text{slope 4: } -V_1 + nV_2 / t \cdot L$$

$$\text{Area: } \frac{2nV_1V_2T_\varphi(T_S/2 - T_\varphi)}{T_S L}$$

Figure 4: Area computation under SPS

Trapezoid 2 (interval II): $S_2 = \frac{(i_\varphi - i_{L,0})(T_S/2 - T_\varphi)}{2}$

Total area is $S = S_1 + S_2$. Expanding and collecting like terms (the $T_\varphi i_\varphi$ terms cancel) gives:

$$S = T_\varphi i_{L,0} + \frac{T_S}{4}(i_\varphi - i_{L,0}) \quad (9)$$

Substituting $i_\varphi - i_{L,0} = (V_1 + nV_2)T_\varphi/L$:

$$S = T_\varphi i_{L,0} + \frac{T_S T_\varphi (V_1 + nV_2)}{4L} \quad (10)$$

Step 5: Substituting back into P_1 and simplifying. Inserting S into (3.10) and splitting into two terms:

$$P_1 = \underbrace{\frac{2V_1T_\varphi}{T_S} i_{L,0}}_{\text{term A}} + \underbrace{\frac{V_1T_\varphi(V_1 + nV_2)}{2L}}_{\text{term B}} \quad (11)$$

Convert all time quantities into angles ($T_\varphi/T_S = \varphi/(2\pi)$), substitute the closed form (3.12) of $i_{L,0}$, and bring everything over the common denominator $4\pi^2 f_S L$:

$$\pi(nV_2 - V_1) + \pi(V_1 + nV_2) = 2\pi nV_2 \quad (\text{the } \pi V_1 \text{ terms cancel completely!}) \quad (12)$$

The bracket simplifies to $2nV_2(\pi - \varphi)$, and substituting back:

DAB single-phase-shift power formula (SPS)

$$P = P_1 = P_2 = \frac{nV_1V_2\varphi(\pi - |\varphi|)}{2\pi^2 f_S L} \quad \forall -\pi < \varphi < \pi \quad (13)$$

The $|\varphi|$ generalizes the positive-phase-shift result to negative phase shift ($\pi - \varphi \rightarrow \pi - |\varphi|$), making the formula valid over full range. $P > 0$ means HV $\rightarrow$ LV; $P < 0$ means LV $\rightarrow$ HV.

Maximum Power and Inverting for the Phase Shift

Maximum power. Setting $\partial P/\partial \varphi = 0$ in (3.13): $\frac{d}{d\varphi}[\varphi(\pi - \varphi)] = \pi - 2\varphi = 0 \Rightarrow \varphi = \pi/2$. Substituting back:

$$|P_{\text{PS,max}}| = \frac{nV_1V_2}{8f_sL} \quad \text{at } \varphi = \pm\pi/2 \quad (14)$$

Read in reverse, maximum power is the design ceiling for L : $L < nV_1V_2/(8f_sP_{\text{required}})$. If L is too large, not even $\varphi = \pi/2$ delivers enough power.

Inverting for φ . Treating (13) as a quadratic equation in φ and applying the quadratic formula gives two roots. Take the root with the minus sign (since $P = 0$ must map to $\varphi = 0$):

$$\varphi = \frac{\pi}{2} \left[1 - \sqrt{1 - \frac{8f_sL|P|}{nV_1V_2}} \right] \text{sgn}(P) \quad \forall |P| < |P_{\text{PS,max}}| \quad (15)$$

The plus-sign root corresponds to $\pi/2 < \varphi < \pi$, where power *decreases* as φ increases; it is not used in practice and is simply discarded. Equation (15) is the SPS control law: given a target power P , it computes φ .

The Problems with SPS

Problem 1: Large circulating current (high conduction loss). Under SPS, $i_{L,0} < 0$ (initial current is negative) while $v_{\text{AC1}} = +V_1$ during interval I. At the beginning of interval I, v_{AC1} and i_L therefore point in opposite directions, so instantaneous power $p_1 = v_{\text{AC1}} \cdot i_L < 0$ and **energy flows back from LV to HV**. This portion of the current “goes out and comes back” without transferring any net power, producing nothing but I^2R loss along the way — hence the name circulating current.

The extreme example given in the book: at $P = 2\text{ kW}$, $V_1 = 450\text{ V}$, $V_2 = 11\text{ V}$, LV-side transformer RMS current reaches 283 A, whereas DC-equivalent current is only $2000/11 \approx 182\text{ A}$. The difference $\sqrt{283^2 - 182^2} \approx 217\text{ A}$ is pure circulating current, and conduction loss is amplified by a factor of $(283/182)^2 \approx 2.4$.

Problem 2: Narrow soft-switching range (high switching loss). This cannot be analyzed with a lossless model; a detailed explanation follows later.

Root cause. SPS locks $D_1 = D_2 = 1/2$, so the only adjustable quantity is φ . But φ controls *when* a square wave starts, not *what the waveform looks like*. New degrees of freedom (D_1 , D_2) must be introduced to solve the problem properly. This is the motivation for triangular and trapezoidal current modes.

Triangular Current Mode Modulation

Design Idea and Objectives

Triangular current mode is defined by $i_L(0) = 0$ and $i_L(t_2) = 0$: current departs from zero, returns to zero, and then rests. This directly brings two benefits.

Benefit 1: Elimination of circulating current. Because $i_L(0) = 0$, current rises from zero (for $\varphi > 0$) and always stays ≥ 0 , never opposing $v_{\text{AC1}} = +V_1$. Hence $p_1 = v_{\text{AC1}} \cdot i_L \geq 0$ holds at all times, and **at no instant does energy flow back from LV to HV**.

Quantitatively versus SPS: SPS has a segment with $p_1 < 0$ (reverse power flow) at the beginning of interval I, whereas triangular mode keeps $p_1 \geq 0$ throughout.

Benefit 2: Zero-current switching (ZCS) on the LV side. LV-side switches commute at $t = t_2$, where $i_L(t_2) = 0$. Therefore $L_{\text{stray}} \cdot di/dt = 0$, no voltage spike appears, and switching loss is zero.

Achieving these objectives requires **releasing the degrees of freedom D_1 and D_2** (no longer locked at $1/2$), which introduces freewheeling states.

Complete Switching-State Table over a Period

Triangular mode has 6 intervals over a complete period (3 in the first half-period, mirrored in the second) and makes use of the freewheeling states of both full bridges. Take $V_1 > nV_2$ with $\varphi > 0$ as an example:

Int.	Time	HV	LV	v_{AC1}	v_{AC2}	v_R	i_L	Segment type
I	$0 \rightarrow t_1$	T_1, T_4 driving	T_5, T_8 driving	$+V_1$	$+V_2$	$V_1 - nV_2$ pos.	$0 \rightarrow \hat{I}_{pk}$ slow rise	Both sides drive
II	$t_1 \rightarrow t_2$	T_1, T_3 freewheel	T_5, T_8 driving	0	$+V_2$	$-nV_2$ neg.	$\hat{I}_{pk} \rightarrow 0$ steep fall	HV freewheels
III	$t_2 \rightarrow T_S/2$	T_1, T_3 freewheel	T_5, T_7 freewheel	0	0	0	stays at 0	Both freewheel
IV	$T_S/2 \rightarrow t'_1$	T_2, T_3 driving	T_6, T_7 driving	$-V_1$	$-V_2$	$-(V_1 - nV_2)$	$0 \rightarrow -\hat{I}_{pk}$	Both sides drive
V	$t'_1 \rightarrow t'_2$	T_2, T_4 freewheel	T_6, T_7 driving	0	$-V_2$	$+nV_2$	$-\hat{I}_{pk} \rightarrow 0$	LV freewheels
VI	$t'_2 \rightarrow T_S$	T_2, T_4 freewheel	T_6, T_8 freewheel	0	0	0	stays at 0	Both freewheel

Table 3: Switching states over one period in triangular current mode

Power Formula and Derivation of the Control Parameters

Exploiting symmetry, we compute the integral of i_L over a half-period; that integral is simply the area of a triangle. Durations T_1 and T_2 determine the power:

$$P = \frac{nV_2}{T_S L} [V_1 T_1 (2T_2 - T_1) - nV_2 T_2^2] \quad \forall V_1 > nV_2 \quad (16)$$

T_2 is fixed by the condition $i_L(t_2) = 0$. Setting current to zero at $t = t_2$:

$$0 = i_L(t_1) - \frac{nV_2}{L} (t_2 - t_1) = \frac{V_1 - nV_2}{L} T_1 - \frac{nV_2}{L} T_2 \quad (17)$$

Rearranging (with $T_1 = t_1$ and $T_2 = t_2 - t_1$):

$$T_2 = T_1 \frac{V_1 - nV_2}{nV_2} \quad (18)$$

The “volt-second area” $(V_1 - nV_2) \cdot T_1$ accumulated during the rising segment must equal the “volt-second area” $nV_2 \cdot T_2$ consumed during the falling segment; otherwise current cannot return to zero.

Relation between phase-shift angle φ and the timings:

$$\varphi = 2\pi f_S \left[\frac{T_1 + T_2}{2} - \frac{T_1}{2} \right] = \pi f_S T_2 \quad (19)$$

Expressing power in terms of φ and f_S . Eliminating T_1 and T_2 gives the closed form:

$$P = \frac{\varphi^2 V_1 (nV_2)^2}{\pi^2 f_S L (V_1 - nV_2)} \quad \forall V_1 > nV_2 \wedge 0 < \varphi < \varphi_{\Delta, a, \max} \quad (20)$$

Compared with SPS: SPS power is quasi-linear in φ (a parabola, $\varphi(\pi - |\varphi|)$), whereas triangular-mode power is $\propto \varphi^2$ (purely quadratic).

Inverting for φ . Taking the square root of (20) directly:

$$\varphi = \pi \sqrt{f_S L P \frac{V_1 - nV_2}{V_1 (nV_2)^2}} \quad \forall V_1 > nV_2 \wedge 0 < \varphi < \varphi_{\Delta,a,\max} \quad (21)$$

Closed forms for T_1 and T_2 .

$$T_1 = \frac{\varphi}{\pi f_S} \frac{nV_2}{V_1 - nV_2} \quad (22)$$

$$T_2 = \frac{\varphi}{\pi f_S} \quad (23)$$

These are all the parameters a controller needs: given P , use (21) to obtain φ , then (22) and (23) to obtain T_1 and T_2 , and switching signals can be generated.

Maximum Power and Constraints

Maximum power. As P increases, T_1 and T_2 grow and the rest segment $T_3 = T_S/2 - T_1 - T_2$ shrinks. When $T_3 = 0$ the triangle fills an entire half-period and power reaches its ceiling:

$$P_{\Delta,a,\max} = \frac{n^2 V_2^2 (V_1 - nV_2)}{4 f_S L V_1} \quad (24)$$

Beyond this value no triangle can be formed, and one must switch to trapezoidal mode.

Corresponding maximum phase-shift angle:

$$\varphi_{\Delta,a,\max} = \frac{\pi}{2} \left(1 - \frac{nV_2}{V_1} \right) \quad (25)$$

Constraint on transformer turns ratio. Requiring $P_{\Delta,a,\max} > 0$ demands $V_1 > nV_2$, that is:

$$n < \frac{V_{1,\min}}{V_{2,\max}} \quad (26)$$

Reverse Power Transfer

When $\varphi < 0$, intervals I and II exchange order and power direction reverses (LV $\rightarrow$ HV). Fig. 5 shows the same modulation scheme, differing only in power direction: the triangle also flips orientation (falling first, then rising). The power formula differs by a single minus sign:

$$P = -\frac{\varphi^2 V_1 (nV_2)^2}{\pi^2 f_S L (V_1 - nV_2)} \quad \forall V_1 > nV_2 \wedge -\varphi_{\Delta,a,\max} < \varphi < 0 \quad (27)$$

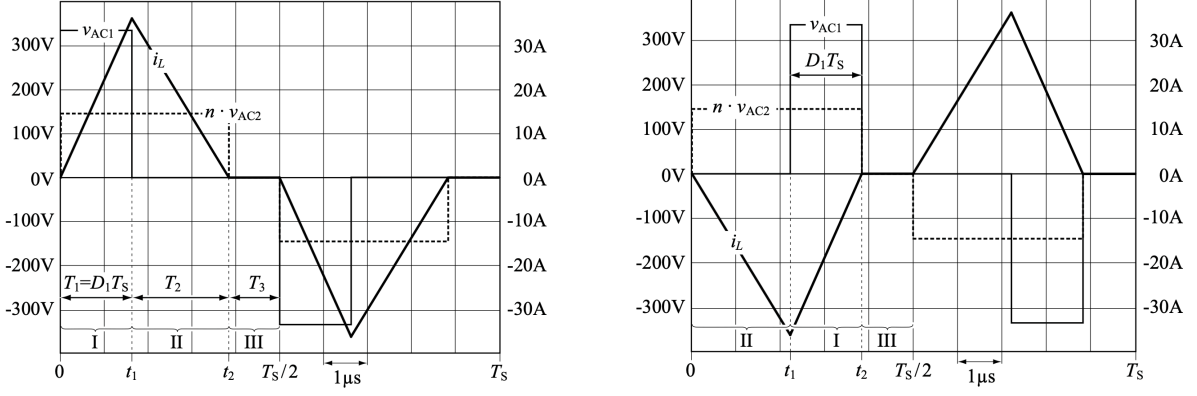


Figure 5: Triangular modulation waveforms

Design Constraint on L

From (24), L must satisfy:

$$L < \min \left(\frac{n^2 V_{2,\min}^2 (V_{1,\min} - n V_{2,\min})}{4 f_S P_{\Delta,a,\max} V_{1,\min}}, \frac{n^2 V_{2,\max}^2 (V_{1,\min} - n V_{2,\max})}{4 f_S P_{\Delta,a,\max} V_{1,\min}} \right) \quad (28)$$

The Case $V_1 < nV_2$

When $V_1 < nV_2$ we have $V_1 - nV_2 < 0$, and the triangular current mode of the previous section (whose denominator $V_1 - nV_2$ turns negative) no longer applies. The book gives another set of triangular-current-mode waveforms, whose power formula is:

$$P = \begin{cases} \frac{\varphi^2 V_1^2 n V_2}{\pi^2 f_S L (n V_2 - V_1)} & \forall V_1 < n V_2 \wedge 0 < \varphi < \varphi_{\Delta,b,\max} \\ -\frac{\varphi^2 V_1^2 n V_2}{\pi^2 f_S L (n V_2 - V_1)} & \forall V_1 < n V_2 \wedge -\varphi_{\Delta,b,\max} < \varphi < 0 \end{cases} \quad (29)$$

Corresponding maximum phase-shift angle:

$$\varphi_{\Delta,b,\max} = \frac{\pi}{2} \left(1 - \frac{V_1}{n V_2} \right) \quad (30)$$

Maximum power combining both voltage relations. Unifying $V_1 > nV_2$ and $V_1 < nV_2$:

$$P_{\Delta,\max} = \begin{cases} \frac{n^2 V_2^2 (V_1 - n V_2)}{4 f_S L V_1} & \text{for } V_1 > n V_2 \\ 0 & \text{for } V_1 = n V_2 \\ \frac{V_1^2 (n V_2 - V_1)}{4 f_S L n V_2} & \text{for } V_1 < n V_2 \end{cases} \quad (31)$$

Switching instant	Side	i_L	Result
$t = 0, T_s/2$	HV	$= 0$	Hard switching ($i_L = 0$ cannot charge/discharge $C_{\text{OSS}} \rightarrow$ ZVS fails)
$t = t_1, t'_1$	HV	$= \hat{I}_{\text{pk}}$	ZVS feasible (sufficient current available)
$t = t_2, t'_2$	LV	$= 0$	ZCS ✓ (zero-current switching, no $L_{\text{stray}} \cdot di/dt$ spike)

Table 4: Soft-switching situation in triangular current mode

Trapezoidal Current Mode Modulation

Motivation: The Power Ceiling of the Triangular Mode

Triangular current mode is limited to $|P| < P_{\Delta, \max}$. When $V_1 \approx nV_2$ (matched voltages), $P_{\Delta, \max}$ approaches zero — precisely the operating condition where efficient operation is most needed is the one where triangular mode cannot be used. Trapezoidal current mode extends the power range by **flattening the “peak” of the triangle into a “plateau”**.

Waveform Features and Interval Structure

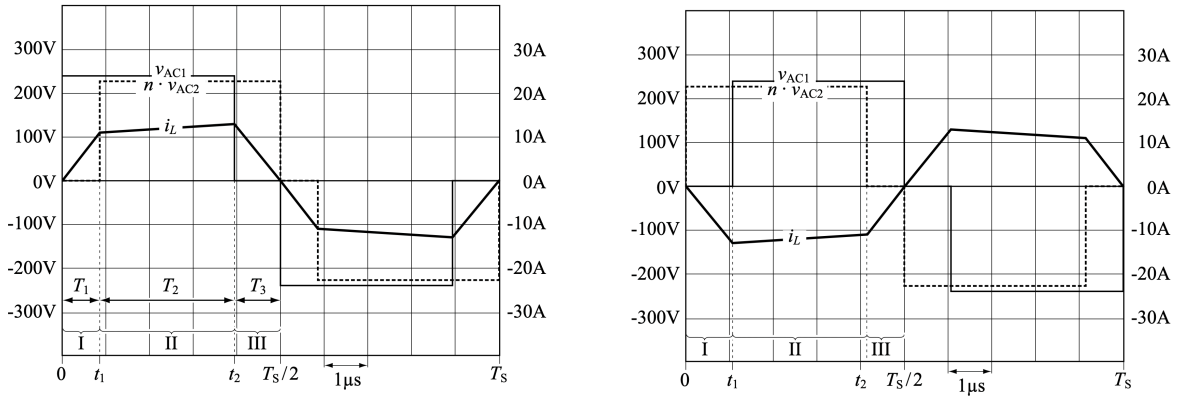


Figure 6: Trapezoidal modulation waveforms

In trapezoidal mode i_L still departs from 0 ($i_L(0) = 0$) and returns to 0 ($i_L(t_2) = 0$), but now with an additional **flat-top plateau** in between. The first half-period splits into 3 intervals:

Int.	Time	HV	LV	v_{AC1}	v_{AC2}	v_R	i_L	Feature
I	$0 \rightarrow t_1$	drive $+V_1$	freewheel 0	$+V_1$	0	$+V_1$ large pos.	$0 \rightarrow$ peak, steep rise	Fast climb
II	$t_1 \rightarrow t_2$	drive $+V_1$	drive $+V_2$	$+V_1$	$+V_2$	$V_1 - nV_2$ small	slow change (plateau)	Push forward
III	$t_2 \rightarrow T_s/2$	freewheel 0	freewheel 0	0	0	0	stays at 0	Rest

Table 5: First half-period of trapezoidal current mode ($V_1 > nV_2$, $P > 0$). The second half-period is mirrored.

Comparison with the Triangular Mode

Int.	Triangular mode		Trapezoidal mode	
	HV/LV	v_R	HV/LV	v_R
I	both drive	$V_1 - nV_2$ (small pos., slow rise)	HV drives + LV freewheels	V_1 (large pos., fast rise)
II	HV freewheels + LV drives	$-nV_2$ (neg., fast fall)	both drive	$V_1 - nV_2$ (small, plateau)
III	both freewheel	0	both freewheel	0

Table 6: Interval-by-interval contrast between triangular and trapezoidal modes

Power Formula

The trapezoidal power formula is more involved than the triangular one:

$$P = \text{sgn}(\varphi) \frac{nV_1V_2 [2nV_1V_2(\pi^2 - 2\varphi^2) - (V_1^2 + (nV_2)^2)(\pi - 2|\varphi|)^2]}{4\pi^2 f_S L (V_1 + nV_2)^2} \quad (32)$$

Maximum power:

$$P_{\square, \max} = \frac{(nV_1V_2)^2}{4f_S L (V_1^2 + nV_1V_2 + (nV_2)^2)} \quad (33)$$

Trapezoid Area > Triangle Area: Extending the Power Range

For the same peak current, a trapezoid (with its plateau) encloses more area than a triangle (with its sharp peak), so trapezoidal mode transfers more power. $P_{\Delta, \max}$ is the ceiling of triangular mode; above it, trapezoidal mode takes over.

Seamless Transition from Triangular to Trapezoidal

Mode-switching strategy

$$\begin{cases} 0 < |P| < P_{\Delta, \max}(V_1, V_2) : & \text{triangular current mode} \\ P_{\Delta, \max} < |P| < P_{\square, \max}(V_1, V_2) : & \text{trapezoidal current mode} \end{cases} \quad (34)$$

At the boundary $|P| = P_{\Delta, \max}$, the sharp peak of the triangle flattens into a plateau of zero width, so both waveforms coincide exactly. In both modes i_L is zero at the instants $t = k \cdot T_S/2$, so the transition happens with **no transient shock** — a seamless handover.

Comparison of the Three Modulation Strategies

Dimension	SPS	Triangular	Trapezoidal
Duty cycle	$D_1=D_2=1/2$	D_1, D_2 free	D_1, D_2 free
Freewheeling state	not used	used	used
$i_L(0)$	$\neq 0$ (can be negative)	$= 0$	$= 0$
Circulating current	substantial	zero	zero
ZCS on LV side	×	✓	✓
ZVS on HV side	partly ✓	×	×
RMS current	highest	lowest	moderate
Control complexity	★ (1 variable)	★★★ (3 variables)	★★★

Table 7: Comparison of SPS, triangular-current, and trapezoidal-current modulation strategies

The evolutionary logic of these three strategies can be summarized as follows.

SPS is the simplest — a single knob φ — but it **cannot change the shape of the current waveform** → large circulating current, high RMS, narrow soft-switching range.

Triangular current mode introduces freewheeling (freeing up D_1 and D_2) and **eliminates circulating current** ($i_L(0) = 0$), sharply reducing RMS → but the HV side loses ZVS, and power hits a ceiling $P_{\Delta, \max}$. The circulating currents of SPS happened to supply exactly the energy needed for ZVS at the switching instants, so eliminating them means accepting hard switching.

Trapezoidal current mode further **extends the power range** (a plateau $\rightarrow$ a larger area), operating between $P_{\Delta,\max}$ and $P_{\square,\max}$, and joins triangular mode seamlessly at the boundary.

The practical engineering strategy is **piecewise switching**: triangular mode at light load (clean and efficient), trapezoidal at medium and heavy load, and SPS as fallback at full load or under extreme wide-voltage conditions (full coverage, lower efficiency). Later chapters of the thesis (Optimal Modulation) go further and use mathematical optimization to find the globally optimal combination of D_1 , D_2 , φ across all operating conditions, without restricting the waveform to a fixed triangular or trapezoidal shape.